AI Playbook and Toolkit for Public Health Departments

AI Risk Appetite and Risk Tolerance for Public Health

GOV 410

Leaders cannot manage AI risk effectively unless they define how much risk the agency is willing to accept for different types of use. Risk appetite describes the broad level of risk an organization is willing to take in pursuit of its mission. Risk tolerance translates that appetite into practical thresholds, conditions, and limits for specific AI uses. This module helps public health leaders define AI risk appetite in a way that reflects public health authority, community trust, legal obligations, health equity, privacy, safety, and operational needs. The goal is not to eliminate all risk. The goal is to make risk decisions explicit, proportionate, documented, and aligned with public health values.

Level	advanced/leadership
Primary track	Governance and Security Track
Appears in tracks	governance-security

Learning Objectives

- Explain the difference between risk appetite, risk tolerance, and risk acceptance.
- Identify AI uses that may be low, moderate, high, or unacceptable risk in a public health context.
- Develop risk tolerance statements for different AI use categories.
- Connect risk tolerance to review level, safeguards, monitoring, and escalation.
- Produce a draft AI risk appetite statement for leadership or governance discussion.

Module Overview

Leaders cannot manage AI risk effectively unless they define how much risk the agency is willing to accept for different types of use. Risk appetite describes the broad level of risk an organization is willing to take in pursuit of its mission. Risk tolerance translates that appetite into practical thresholds, conditions, and limits for specific AI uses.

This module helps public health leaders define AI risk appetite in a way that reflects public health authority, community trust, legal obligations, health equity, privacy, safety, and operational needs. The goal is not to eliminate all risk. The goal is to make risk decisions explicit, proportionate, documented, and aligned with public health values.

Jurisdiction and Agency Policy Note

This module provides national-level training guidance for public health agencies and partners. It is not legal advice and does not replace review by agency legal counsel, privacy officers, procurement officials, civil rights staff, records officers, or other authorized decision-makers. Requirements may vary by state, locality, territory, Tribe, agency, funding source, data-sharing agreement, emergency authority, and organizational policy. Learners should use this module to identify the issues that require review and should confirm applicable requirements before approving, procuring, deploying, or publicly communicating about AI-supported work.

Why This Topic Matters for Public Health AI

Public health agencies make risk decisions every day, but AI can make those decisions less visible. A generative AI tool might be acceptable for drafting internal meeting notes but not for producing unsupervised public health guidance. A predictive model might be acceptable for planning staffing levels but not for denying services. An agentic workflow might be acceptable for creating a draft task but not for sending an enforcement notice without human approval.

Risk appetite helps agencies explain these distinctions. It provides a leadership-level statement of where the agency encourages innovation, where it requires caution, and where it prohibits use. It also helps staff understand that responsible AI is not a binary choice between adoption and avoidance. It is a disciplined process for matching safeguards to risk.

Public Health Example

An agency may define a higher risk appetite for internal administrative uses that do not involve sensitive data, public communication, or public health action. It may define a moderate appetite for staff-facing decision support when human review, validation, and documentation are required. It may define a very low appetite for unsupervised public-facing advice, automated eligibility decisions, enforcement actions, or tools that process sensitive data without approved safeguards.

These statements help governance reviewers make consistent decisions. Instead of debating each proposal from the beginning, reviewers can ask whether the proposed use fits the agency's defined risk category and whether required safeguards are present. The discussion becomes more disciplined and less dependent on individual comfort with AI.

Technical and Governance Deep Dive

Risk appetite should be expressed in plain language and translated into operating rules. For example, an agency might state that it encourages low-risk AI uses that reduce internal administrative burden when no sensitive data are used and outputs are reviewed by staff. It might state that it will consider moderate-risk uses involving operational data only with documented validation, access controls, human review, and monitoring. It might state that it does not permit AI to make final public health enforcement, eligibility, or legal decisions without authorized human review.

Risk tolerance turns these statements into thresholds. A tolerance may specify that a tool cannot be launched if source traceability is absent, if subgroup performance cannot be assessed for a prioritization model, if the false negative rate exceeds a defined threshold, or if the vendor cannot provide adequate documentation. Tolerances should be appropriate to the use case. A lower threshold may be acceptable for internal drafting than for case triage.

Risk appetite should be reviewed regularly. Public expectations, laws, model capabilities, agency experience, and technical safeguards change over time. A use that was too risky during the first year of AI adoption may become acceptable after better infrastructure, policies, and monitoring are in place. Conversely, a use may become unacceptable if incidents, evidence, or community concerns reveal new harms.

Risks, Failure Modes, and Guardrails

A common failure mode is vague risk language. Phrases like 'use AI responsibly' do not help staff decide what is allowed. A guardrail is to define concrete examples of encouraged, conditional, restricted, and prohibited uses.

Another risk is risk appetite that ignores equity. A tool may appear operationally useful while shifting burden to communities that are already underrepresented or over-surveilled. A guardrail is to require equity impact review for uses that affect prioritization, access, enforcement, communication, or resource allocation.

Application to AI-Supported Workflows

Risk appetite applies differently across AI types. Generative AI for internal drafting may fall into a lower-risk category when reviewed by humans. Predictive models used for prioritization require stronger validation and monitoring. Agentic tools that can take system actions require strict limits, stop authority, and auditability. Risk appetite helps define these boundaries in advance.

Reflection Questions

Which current or proposed AI use case in your agency raises this leadership or governance issue most clearly?

Who currently has authority to make decisions about this issue, and is that authority documented?

What evidence would governance or leadership need before approving the use case?

Which staff, partners, or communities would be affected if this issue were handled poorly?

What policy, process, or artifact should be created or updated after this module?

Practical Exercise

Draft a risk appetite statement and a risk tolerance table for at least four AI use categories: internal productivity, staff decision support, public-facing communication, and automated or agentic workflows.

Before You Begin

Choose a real or realistic public health AI use case. Document assumptions, unresolved questions, and which agency roles would need to review the artifact before it is adopted. The exercise should produce a work product that could support governance review, leadership discussion, implementation planning, or policy development.

Expected Artifact or Evidence

AI risk appetite statement; risk tolerance table; prohibited-use examples; mitigation and escalation conditions.

Expected Assignment

- AI risk appetite statement; risk tolerance table; prohibited-use examples; mitigation and escalation conditions.

Knowledge Check

- 1. What is the strongest reason this module topic belongs in AI leadership and governance training?
- 2. Which practice best supports accountable public health AI governance?
- 3. When should an AI use case be escalated for leadership or governance review?
- 4. What is weak evidence of completion for a leadership and governance module?
- 5. What should leaders do when an AI use case has meaningful benefits but unresolved risks?

References and Resources

- NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0): https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- NIST Privacy Framework
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- WHO Ethics and Governance of Artificial Intelligence for Health: https://www.who.int/publications/i/item/9789240029200
- OMB Memorandum M-24-10 on federal AI governance, risk management, and innovation, where relevant to public-sector governance models
- Agency strategic plans, procurement policies, privacy policies, cybersecurity policies, records policies, and equity plans